SPACE: Style-Preserving Attribute-Constrained Erasure for Artist-Style Unlearning in Text-to-Image Diffusion Models

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Genμ 2.0 Challenge — Track 3: Visual Concept Unlearning

github.com/Vedang-P/genmu2026-space-submission

Abstract. We present SPACE (Style-Preserving Attribute-Constrained Erasure), a preservation-first method for erasing artist styles from Stable Diffusion v1.4. Rather than treating erasure strength and content/quality preservation as a single objective to be traded off, SPACE edits a frozen base model with saliency-selected LoRA adapters trained against a dual-anchor teacher that simultaneously anchors toward neutral, content-preserving output and explicitly subtracts an estimated target-style residual. We validate SPACE on two independent target concepts — Vincent van Gogh (submission target) and Thomas Kinkade (independent replication) — using a standardized protocol of 500 matched target-prompt pairs and a full 30,000-image FID evaluation against the official MS-COCO 2014 reference set. SPACE reaches FID within 0.07–0.08 of the unedited base model (14.57 vs. 14.50 for Van Gogh; 14.58 vs. 14.50 for Thomas Kinkade) while measurably suppressing target style (style-score drop 0.045–0.064, target-style hit-rate 32–48%), showing that broad generative utility and targeted style suppression are jointly achievable rather than mutually exclusive.

1 Introduction

Text-to-image diffusion models memorize and reproduce recognizable artist styles, raising copyright and consent concerns and motivating concept erasure (machine unlearning) methods. Prior approaches make different trade-offs: ESD [1] performs a gradient-based edit that moves target-concept predictions away from the original, but erasure strength can trade hard against semantic content and image quality; UCE [2] solves a closed-form linear edit over erased and preserved concepts, but is not trajectory-aware and treats each target prompt as a single point rather than a factorized content/style pair; Concept Ablation [3] maps a target concept toward a generic anchor distribution, which produces visually pleasing results but can under-erase and damage nearby concepts.

SPACE's premise is that erasure and preservation should be jointly engineered, not opposing forces in one loss term. It factorizes each target prompt into content, target-style, neutral-art, and generic-image views; builds a teacher that anchors toward neutral content and explicitly subtracts the estimated style residual; edits only a saliency-selected, low-rank subset of cross-attention projections; supervises across multiple denoising trajectory bands; and replays both general prompts and CLIP-mined vulnerable nearby styles during training to bound collateral damage.

2 Method

Prompt views. For a target prompt p_t (e.g. "Bedroom in Arles by Vincent van Gogh"), SPACE constructs a content-only prompt p_c (artist phrase stripped), a neutral art anchor p_a ("a high quality painting of {content}"), and a generic image anchor p_g ("a high quality image of {content}"), plus robust paraphrase variants of p_t ("in the style of", "inspired by", "as painted by") so the edit is not tied to one literal surface form.

Teacher construction. Let ϵ_θ be the frozen base U-Net and x_t a noisy latent at timestep t. The SPACE teacher is

$$\begin{aligned} n &= (1-\alpha)\,\epsilon_\theta(x_t,\tau(p_c)) + \alpha\,\epsilon_\theta(x_t,\tau(p_a)) \\ d &= \epsilon_\theta(x_t,\tau(p_t)) - \epsilon_\theta(x_t,\tau(p_c)) \\ \text{teacher} &= n - \beta\, d, \end{aligned} \tag{1}$$

with α=0.30 and β=1.0 by default. Eq. 1 both anchors the target prompt toward a neutral, content-preserving prediction and explicitly pushes away from the target-style direction d, rather than only imitating a generic replacement distribution.

Student. The base U-Net is frozen; SPACE trains only LoRA adapters (rank 8) inserted into a saliency-selected subset of cross-attention linear projections. Saliency is measured by probe-gradient norm on a small target/band subset, and the top quartile of candidate modules is edited — the edited set is data-dependent, not a fixed layer list. The "teacher" is the same U-Net with adapters disabled, so no separate frozen expert network is required.

Loss. Training supervises three trajectory bands ($t \in \{0.2, 0.5, 0.8\}$) with

$$\mathcal{L} = \mathcal{L}_{\text{erase}} + \lambda_s\mathcal{L}_{\text{style}} + \lambda_c\mathcal{L}_{\text{content}} + \lambda_i\mathcal{L}_{\text{image}} + \lambda_p\mathcal{L}_{\text{preserve}} + \lambda_v\mathcal{L}_{\text{vuln}} + \lambda_r\mathcal{L}_{\text{lora}},$$

where $\mathcal{L}_{\text{erase}}$ matches the student's target-prompt prediction to the teacher; $\mathcal{L}_{\text{style}}$ penalizes projected energy onto a rank-4 style subspace built by SVD over several target-prompt variants (not a single residual vector); $\mathcal{L}_{\text{content}}/\mathcal{L}_{\text{image}}$ anchor p_c/p_g to the frozen base model; $\mathcal{L}_{\text{preserve}}$ replays general prompts from a 30k COCO pool on an independently sampled latent; $\mathcal{L}_{\text{vuln}}$ replays CLIP-mined nearby artist styles most at risk of collateral damage; and $\mathcal{L}_{\text{lora}}$ regularizes adapter magnitude. Training uses a two-stage curriculum (300 narrow-supervision steps, then 100 steps with broader prompt-family and style-basis coverage), learning rate 10^{-4}.

3 Experimental Setup

Base model. CompVis Stable Diffusion v1.4, edited in place; no architecture changes.

Target concepts. Vincent van Gogh (submission target) and Thomas Kinkade (independent second concept, identical protocol), each trained and evaluated separately.

Protocol. 50 target prompts $\times$ 10 samples = 500 matched vanilla/SPACE pairs per concept, plus 30,000 SPACE generations over a fixed COCO prompt/seed set, scored by canonical torch-fidelity FID against the official MS-COCO 2014 validation reference — the same protocol ESD [1] reports vanilla SD v1.4 FID (14.50) under.

Baselines. During development, SPACE was benchmarked against ESD-x, UCE, and Concept Ablation on a shared 50-prompt Van Gogh set using official upstream implementations (not reimplementations), all released in the accompanying repository.

Metrics. FID (general utility); CLIP image similarity and DINO similarity (edited-vs-vanilla structural/content agreement); CLIP style score drop and target rate (fraction of forget-set samples still hitting the target style — erasure strength); LPIPS (perceptual distance).

4 Results

Table 1 reports both concepts under the identical protocol. FID stays within 0.07–0.08 of the unedited model at 30k-image scale for both concepts, indicating general image quality and prompt-following ability on unrelated content are essentially undamaged. At the same time, target-style hit rate drops to 32.4% (Van Gogh) and 48.4% (Thomas Kinkade) with a corresponding CLIP style-score drop, showing real, measurable suppression of the target style rather than a null edit.

Metric	Van Gogh	T. Kinkade
FID (30k, canonical)	14.57	14.58
vs. vanilla SD v1.4 [1]	14.50	14.50
CLIP image similarity ↑	0.798	0.755
DINO similarity ↑	0.622	0.471
Style target rate ↓	0.324	0.484
Style score drop ↑	0.045	0.064
LPIPS (vs. vanilla)	0.569	0.578

Table 1: SPACE results, 500 matched pairs + 30k-image COCO FID per concept. ↑/↓: higher/lower is better for erasure-preservation trade-off.

That the same hyperparameters transfer across two stylistically distinct artists (Van Gogh's post-impressionist brushwork vs. Kinkade's luminist realism) without per-concept retuning supports that the erasure-preservation trade-off is a property of the method, not an artifact of one target.

Qualitative. Fig. 1 shows representative pairs: SPACE consistently removes the signature stylistic cue (Van Gogh's impasto brushwork; Kinkade's diffuse warm-light glow) while retaining scene layout and subject identity. The full 500-pair grids and per-concept metric dashboards are released with the code.

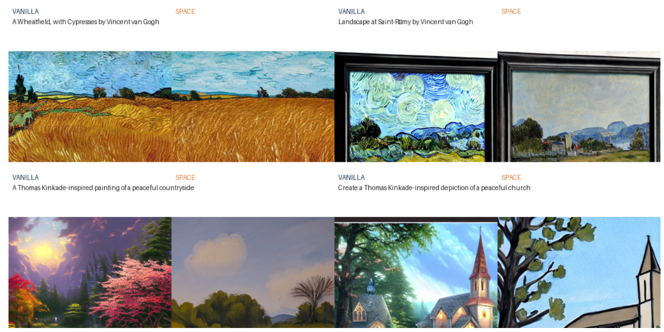

Figure 1: Vanilla SD v1.4 vs. SPACE, matched seed/prompt. Top: Van Gogh. Bottom: Thomas Kinkade.

5 Discussion and Limitations

A minority of Van Gogh prompts (e.g. "The Reaper", "Cypresses") drift toward a recurring dark-robed-figure motif rather than a close content match, indicating the content anchor does not perfectly constrain every prompt and some concept-specific failure modes remain. Style target rate (32–48%) shows suppression is substantial but not absolute; fully adversarial or indirect phrasings were not exhaustively probed in this evaluation and are a natural next robustness check. Finally, SPACE targets output-level style suppression under this evaluation harness — it is not a formal proof that all target-concept information is removed from model weights, and should not be read as such.

6 Conclusion

SPACE combines a dual-anchor, style-residual-subtracting teacher with localized, saliency-selected LoRA adapters and multi-axis preservation replay to erase artist style while keeping FID within a fraction of a point of the unedited model at COCO scale. Validated identically on two independent concepts, code, weights, and generated samples (forget and retain sets) are released for reproduction at the repository linked above.

References

[1] Gandikota, R., Materzynska, J., Fiotto-Kaufman, J., Bau, D. Erasing Concepts from Diffusion Models. ICCV, 2023.

[2] Gandikota, R., Orgad, H., Belinkov, Y., Materzynska, J., Bau, D. Unified Concept Editing in Diffusion Models. WACV, 2024.

[3] Kumari, N., Zhang, B., Wang, S.-Y., Shechtman, E., Zhang, R., Zhu, J.-Y. Ablating Concepts in Text-to-Image Diffusion Models. ICCV, 2023.